



GROUP 28 · CSIT321

Usability Evaluation

Certacito.ai · A4 Prototype Presentation appendix

Project Certacito.ai, AI agent governance platform

Document usability-evaluation.md

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Live system <http://20.92.93.30>

A4 appendix. The A3 marking feedback asked for "additional user testing or usability evaluation", so this document records the evaluation work done for A4 and is honest about its limits. In short: we ran a structured expert review of the live deployment (a heuristic walkthrough plus a WCAG 2.1 AA audit, run against the running Azure system on 7 August 2026 with the simulator on), and we collected feedback from the real people who saw the system along the way. We did not run formal moderated sessions with recruited participants - section 5 covers why and what we would do about it in A5.

1. Method

Three sources of evidence, in order of rigour:

- Heuristic walkthrough.** Both persona journeys from A3 (Compliance Officer triage, DevSecOps configuration) walked end-to-end on the live deployment against Nielsen's ten heuristics. Every screen visited, every primary action exercised, including failure paths (wrong password, empty states).
- WCAG 2.1 AA audit.** Contrast ratios computed for every colour pair in the branding style guide using the WCAG relative-luminance formula, plus scripted checks on the live site (Playwright): keyboard-only navigation of the login, focus indicator visibility, error-state messaging, and table semantics for screen readers.
- Collected feedback.** The unprompted reactions we already had from real viewers: the sponsor's A3 review, the supervisor's June email, and visitors at the project exhibition. These are catalogued in section 4.

2. Issues found

The audit found nine issues. Four were fixed before submission; five remain open with concrete recommended fixes. Severity is our judgement of user impact.

ID	FINDING	HEURISTIC / CRITERION	SEVERITY	STATUS
U-01	Sign-in crashed on a fresh build (a component read a user value that was never passed to it). Found while preparing this evaluation.	Error prevention	High	Fixed - and a TypeScript check was added to CI so this class of bug fails the build instead of shipping
U-02	Agent registry showed "0 of 0 agents" while agents were live and reporting	Visibility of system status	Medium	Fixed - registry now maps live API data directly
U-03	Compliance score displayed 100% before any traffic existed, which reads as a (false) claim	Match with the real world	Medium	Fixed - shows a dash until there is data to score
U-04	Panel titled "Assigned policy rules" actually listed permitted action types	Match with the real world	Low	Fixed - renamed
U-05	Permit-green text (#27AE60) on white measures 2.87:1, below	WCAG 1.4.3 contrast	Medium	Open - recommended #1E8449 (4.72:1). Partly mitigated by the design

	the 4.5:1 AA minimum for normal text			rule that outcome is never conveyed by colour alone
U-06	Escalate-amber text (#E67E22) on white measures 2.85:1	WCAG 1.4.3 contrast	Medium	Open - recommended #9C640C (4.95:1)
U-07	Muted slate text (#6B7A99) on white measures 4.31:1 - passes for large text, marginally fails for normal text	WCAG 1.4.3 contrast	Low	Open - recommended #5A6B8C (5.36:1)
U-08	The email and password inputs suppress the browser focus outline without providing a replacement, so a keyboard user cannot see which field has focus. Buttons are unaffected (they keep the default outline)	WCAG 2.4.7 focus visible	Medium	Open - restore a visible focus ring on text inputs
U-09	No <code>aria-label</code> anywhere in the console; icon-only controls such as the notification bell expose only their badge number to a screen reader	WCAG 4.1.2 name, role, value	Medium	Open - label icon-only buttons

3. What passed

Worth recording, because these were deliberate A3 design decisions and the audit confirms they held up in the implementation:

- **Core palette contrast.** Navy on white 11.27:1, teal on white 5.62:1, red on white 5.44:1, and the same pairs reversed on dark backgrounds - all pass AA for normal text.
- **Keyboard path through login.** Tab order is logical (email, password, sign in, SSO, back to site) and everything is reachable without a mouse.
- **Error messaging.** A wrong password shows "Invalid email or password" - a clear recovery message that deliberately does not reveal which half was wrong.
- **Honest placeholder.** The SSO button is labelled "Sign in with SSO (Disabled)", so it cannot be mistaken for a broken control.
- **Table semantics.** The audit log renders as a real `<table>` with proper header cells, so screen readers get column context for free.
- **Risk is never colour-alone.** Every risk and outcome badge pairs colour with a text label, which is what keeps U-05/U-06 at Medium rather than High.

4. Feedback from real viewers

We did not have access to a pool of enterprise compliance officers, so the feedback below is opportunistic rather than sampled - but all of it changed the product, which is the point of collecting it:

SOURCE	FEEDBACK	WHAT WE CHANGED
Anthony Autore (sponsor), A3 review	Lead with interception, not inventory	Dashboard now opens on the live intercepted-actions feed

Supervisor email, June	A new viewer should understand the system in a few minutes	Public landing page plus the self-contained healthcare demo scenario
Exhibition visitors	Wanted the "what is this and why should I care" story before seeing a console	The landing page exists because of this

5. Limitations, honestly

- **No moderated user testing.** We did not run task-based sessions with recruited participants, and we did not administer SUS - with only the team as a sample the numbers would be noise, not evidence. The right participants are the sponsor's compliance staff; sessions with them are the first item we would schedule in A5.
- **Screen reader coverage is partial.** We audited the semantics (tables, labels, names) programmatically but did not do a full NVDA/VoiceOver run-through.
- **Contrast audit scope.** We measured every pair documented in the branding style guide; ad hoc colours in individual components were not exhaustively swept.

6. Summary

The implemented console holds up well against the heuristics it was designed to: status is visible everywhere, errors are recoverable and clearly worded, and the structural palette is comfortably AA-compliant. The audit still earned its keep - it caught a login-breaking bug before submission, plus four fixed and five open issues, each open one with a tested replacement value ready to apply. The honest gap is moderated testing with real compliance users, which needs access we do not have this session and is flagged for A5.